



Quantum-inspired dynamical models on quantum and classical annealers

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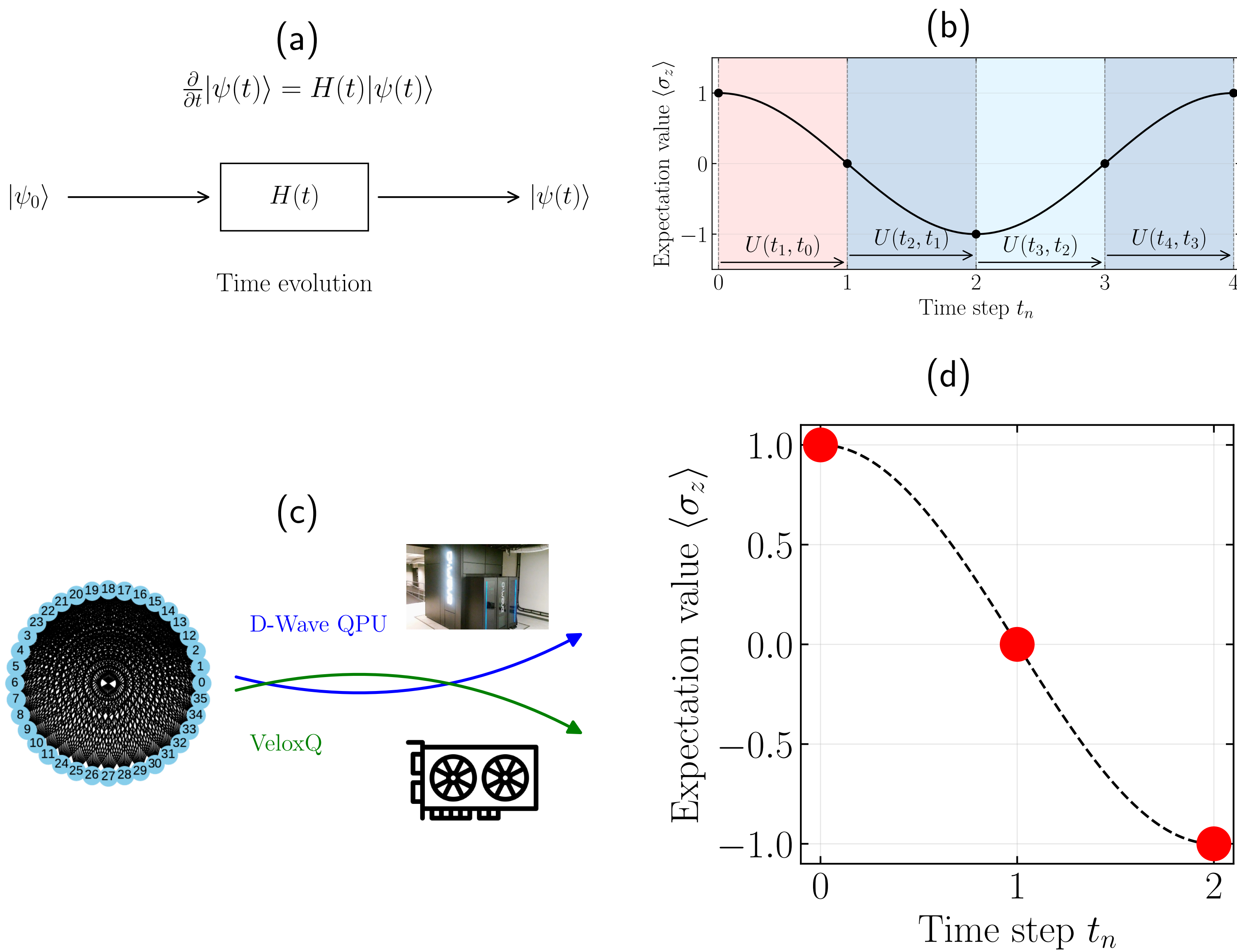
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Introduction

We propose a practical, physics-inspired benchmarking suite to challenge both quantum and classical computers by mapping real-time quantum dynamics to a common optimization format.[1]



Problem Description

- ▶ Quantum State Evolution: $|\Psi\rangle = \sum_{n=0}^{N-1} |t_n\rangle \otimes |\psi(t_n)\rangle$.
- ▶ Annihilated by Hermitian clock operator[4]

$$\mathcal{C} = \sum_{n=0}^{N-2} \left(|t_{n+1}\rangle \langle t_{n+1}| \otimes I - |t_{n+1}\rangle \langle t_n| \otimes U_n + \text{h.c.} \right)$$

$$\min f(\mathbf{x}) = \frac{1}{2} \langle \mathbf{x} | \mathcal{A} | \mathbf{x} \rangle - \langle \mathbf{x} | \phi \rangle$$

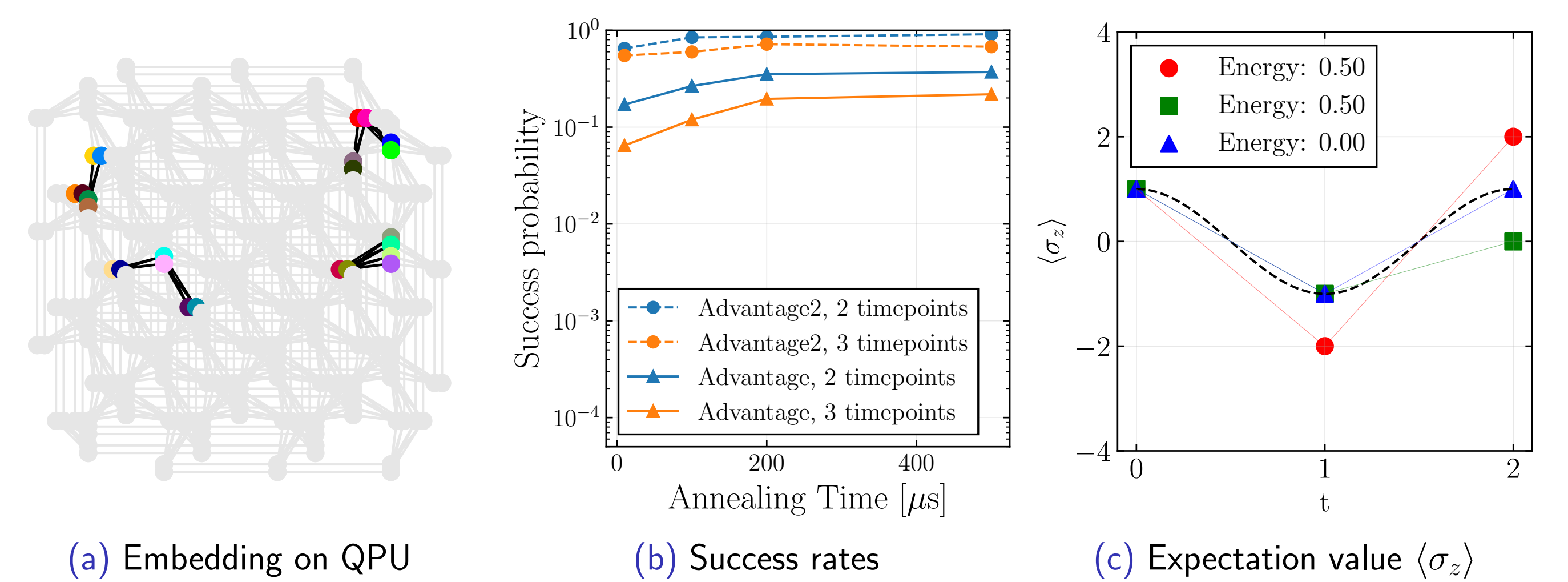
$$\mathcal{A} := \mathcal{C} + |t_0\rangle \langle t_0| \otimes I$$

$2\hat{I}$	$\hat{U}^\dagger(t_1, t_0)$	0	0
$\hat{U}(t_1, t_0)$	$2\hat{I}$	$\hat{U}^\dagger(t_2, t_1)$	0
0	$\hat{U}(t_2, t_1)$	$2\hat{I}$	$\hat{U}^\dagger(t_3, t_2)$
0	0	$\hat{U}(t_3, t_2)$	$\hat{I}$

Clock Hamiltonian $\mathcal{C}$

Solving process

- ▶ Example System 1 - Single qubit Y -rotation $H_1 = \frac{\pi}{2} \sigma_y$,



Methods

Quantum Annealing

- ▶ Driver Hamiltonian:

$$|+\rangle^{\otimes N} = \frac{1}{\sqrt{2^N}} \sum_{x \in \{0,1\}^N} |x\rangle$$

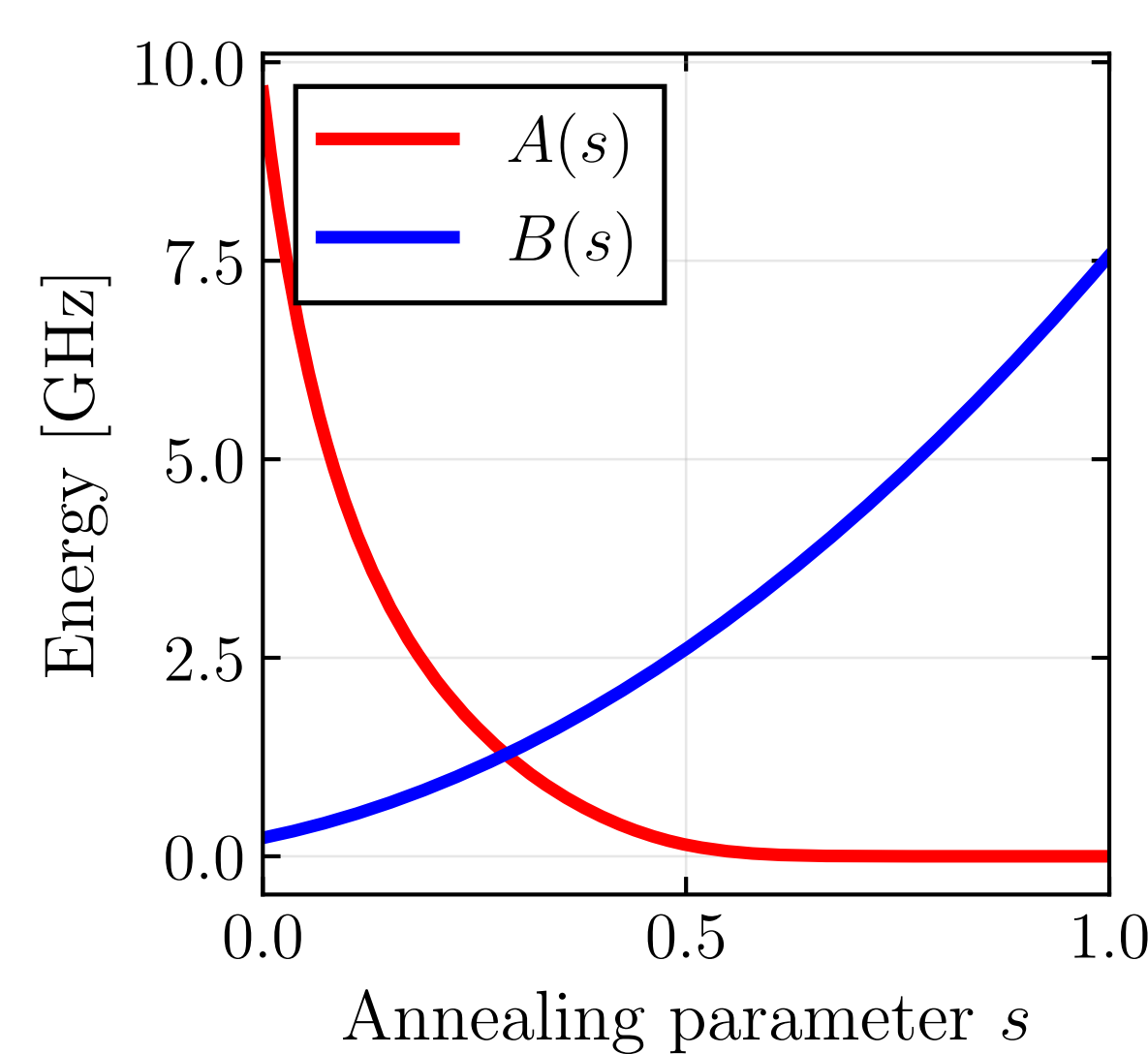
where N is the number of qubits.

- ▶ Problem Hamiltonian:

$$H_P = \sum_{i=1}^N h_i \hat{\sigma}_z^{(i)} + \sum_{i<j} J_{ij} \hat{\sigma}_z^{(i)} \hat{\sigma}_z^{(j)}.$$

- ▶ System evolution:

$$H_{\text{Ising}} = \frac{A(s)}{2} H_D + \frac{B(s)}{2} H_P.$$

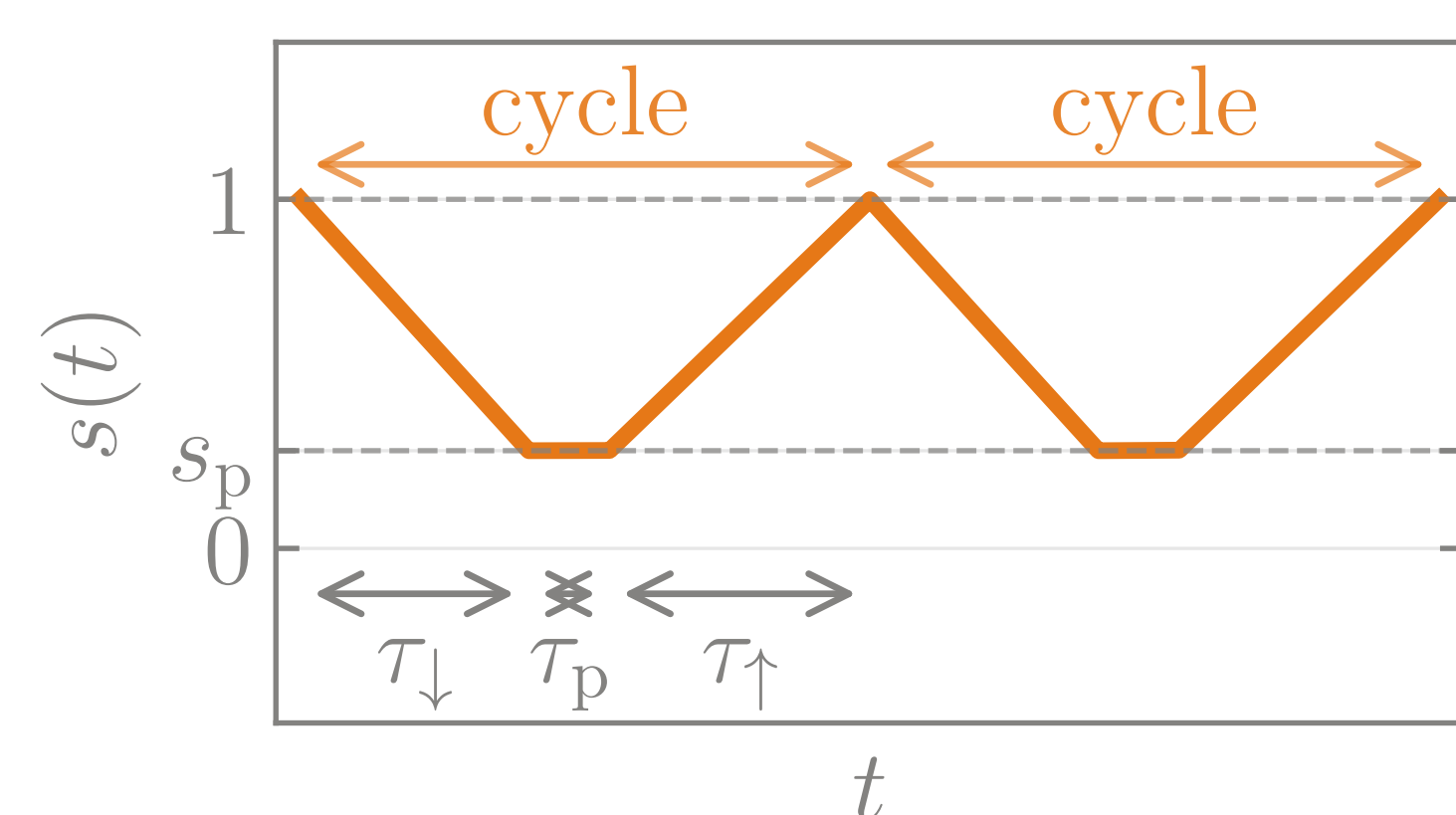


VeloxQ

- ▶ Physics-inspired QUBO (quadratic unconstrained binary optimization) solver

Cyclic Annealing [2, 3]

- ▶ soften the landscape around a candidate solution and re-freeze it
- ▶ reuse promising configuration for next cycle

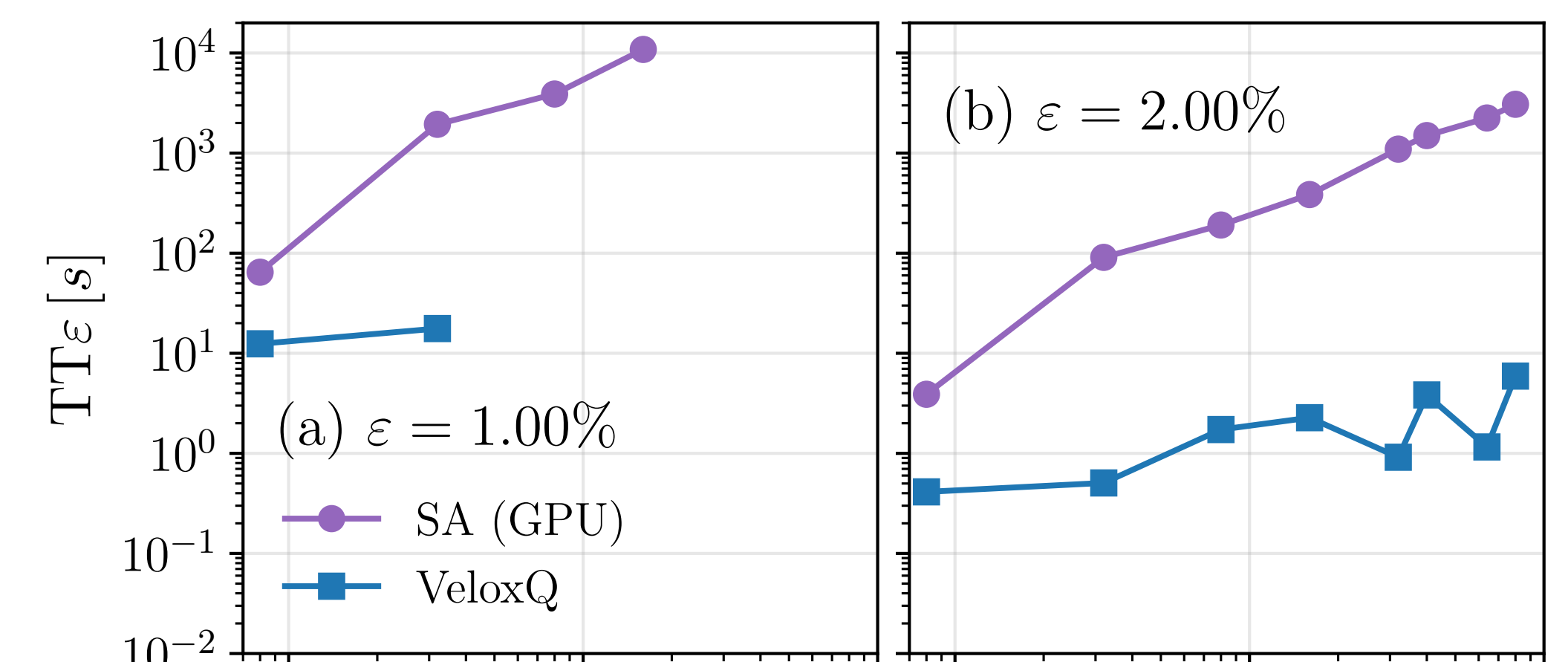


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Results

Time-to-epsilon on large instances



TTE scaling for VeloxQ and SA (GPU) on large QUBO instances (N up to 10^5 , systems H_1, H_2, H_6). VeloxQ outperforms SA for most targets $\epsilon \in \{1.0, 2.0, 3.5, 5.0\}\%$.

System-resolved TTS₉₉ scaling

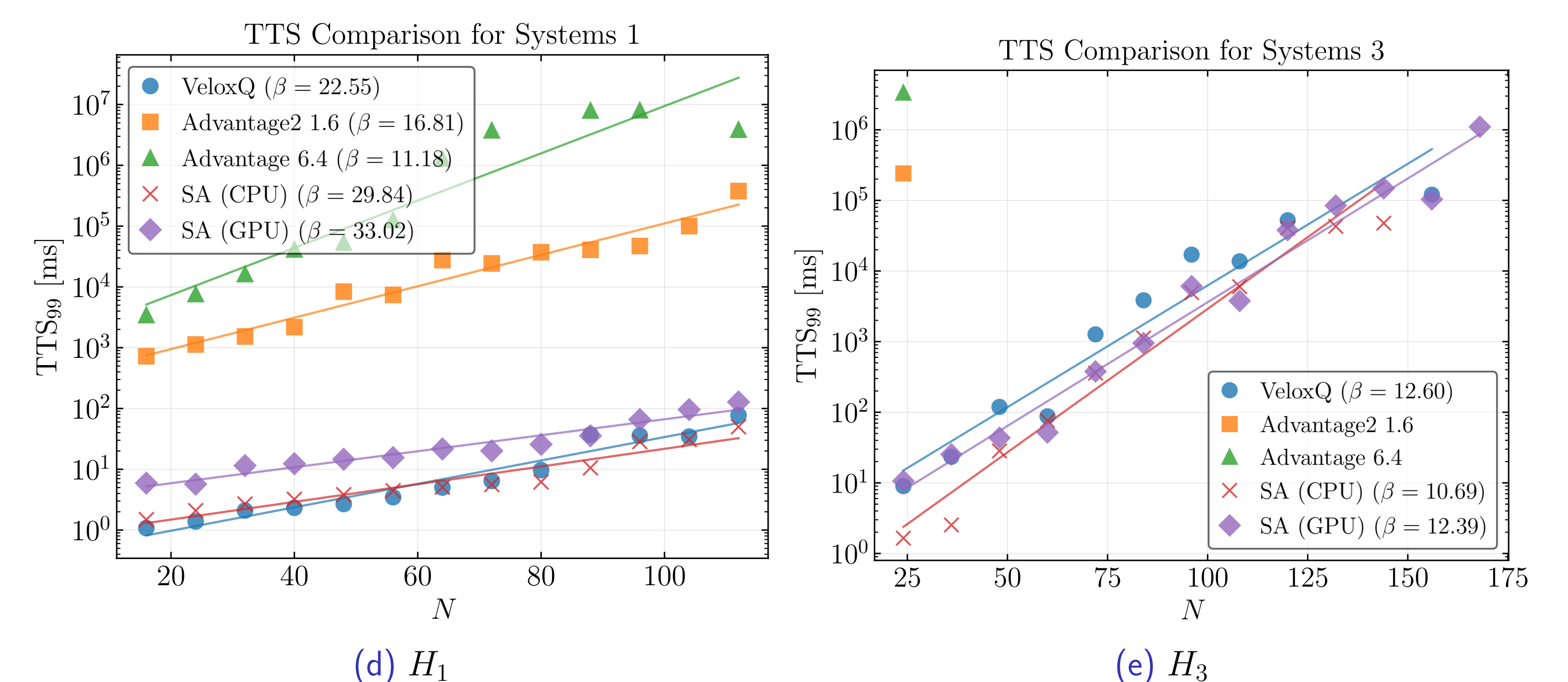


Figure: TTS₉₉ vs. N : exponential scaling $\propto \exp(N/\beta)$. GPU solvers dominate for $N \gtrsim 100$; missing points indicate zero success probability.

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